MATH 101 Unit 2: Functions and Their Graphs - Practice 1

Version 202610122020.37 | Name: _____

Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used. Suggested timed practice: 30 minutes.

Answer key follows the answer sheet. Attempt the test before checking it.

1. (a) Choose the pair giving h(x) = f(g(x)). (b) Form F(G(x)) and evaluate it at the supplied input.

$h(x) = (2x + 1)^2 + 3$

$A.\quad f(x) = 2x + 1,\ g(x) = x^2 + 3$

$B.\quad f(x) = 2x^2 + 3,\ g(x) = x + 1$

$C.\quad f(x) = x^2 + 1,\ g(x) = 2x + 3$

$D.\quad f(x) = x^2 + 3,\ g(x) = 2x + 1$

$(b)\quad F(x) = x^2,\ G(x) = x - 2,\ x_0 = 3$

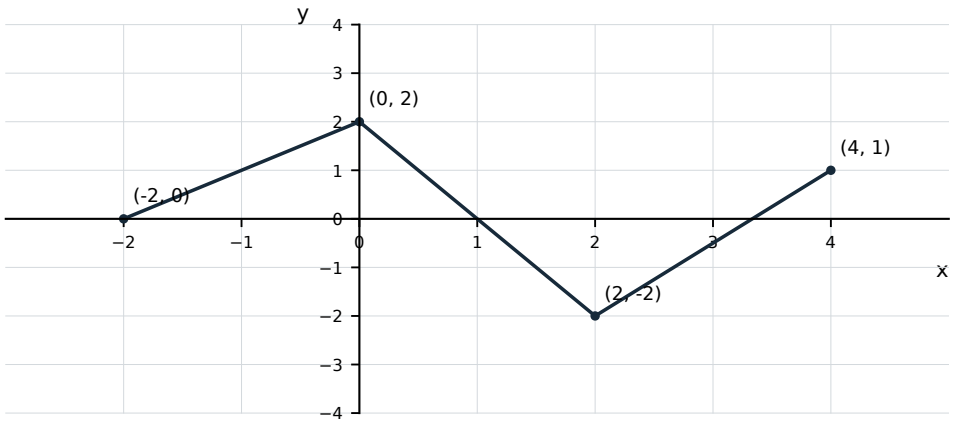
2. For this function, find (a) f(-1), (b) domain, and (c) range.

$f(x) = 2|x - 1| - 2$

3. (a) Find the linear function through these points, with domain all real numbers. (b) Find its inverse and give the inverse domain and range.

$(1,\ 1);\ (3,\ 5)$

4. From the graph, find (a) average rate from x = -2 to x = 4; (b) increasing and decreasing intervals; (c) the two local extrema (ordered pairs).



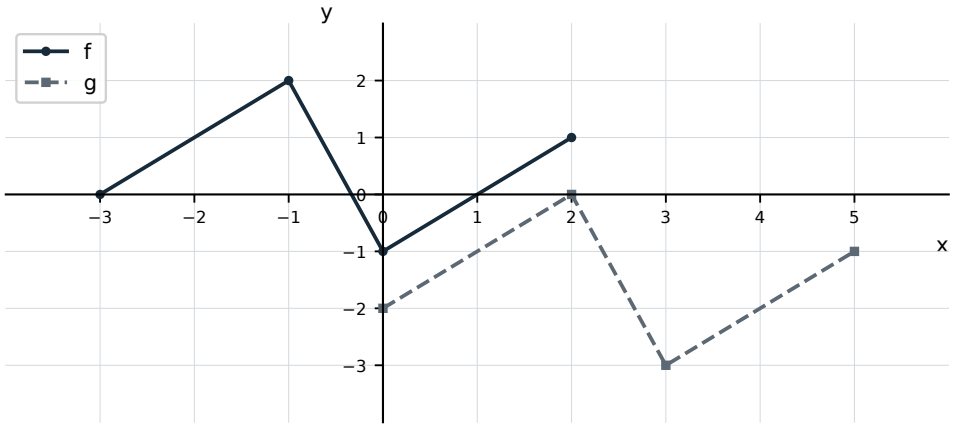
5. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = f(x-3) - 2$

B. $g(x) = f(x+3) + 2$

C. $g(x) = f(x+3) - 2$

D. $g(x) = f(x-3) + 2$



Student answer sheet

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Question 1

(a) Choice

8 pts

(b) Composition

8 pts

(b) Value

4 pts

Question 2

(a) Value

6 pts

(b) Domain

7 pts

(c) Range

7 pts

Student answer sheet

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Question 3

(a) Function

10 pts

(b) Inverse

6 pts

(b) Domain

2 pts

(b) Range

2 pts

Question 4

(a) Rate

6 pts

(b) Increasing

4 pts

(b) Decreasing

4 pts

(c) Local max

3 pts

(c) Local min

3 pts

Question 5

Choice

20 pts

Answer key

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Question 1

(a) Choice	D	8 pts
(b) Composition	$x^2 - 4x + 4$	8 pts
(b) Value	1	4 pts

Question 2

(a) Value	2	6 pts
(b) Domain	$(-\infty, \infty)$	7 pts
(c) Range	$[-2, \infty)$	7 pts

Answer key

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Question 3

(a) Function	$f(x) = 2x - 1$	10 pts
(b) Inverse	$f^{-1}(x) = \frac{x}{2} + \frac{1}{2}$	6 pts
(b) Domain	$(-\infty, \infty)$	2 pts
(b) Range	$(-\infty, \infty)$	2 pts

Question 4

(a) Rate	$\frac{1}{6}$	6 pts
(b) Increasing	$(-2, 0) \cup (2, 4)$	4 pts
(b) Decreasing	$(0, 2)$	4 pts
(c) Local max	$(0,\ 2)$	3 pts
(c) Local min	$(2,\ -2)$	3 pts

Question 5

Choice	A	20 pts